AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM356
Interaction Design and User Experience

Final Examination
Spring Semester 2025/2026

Date: 09 /June/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: Multiple Choice questions [10 x 2 = 20 Marks]

Part 1: ANSWER ALL QUESTIONS [20 marks]

1. Which term describes a high-level description of how a system is organized and operated, outlining what people can do with a product?
   a) Mental model b) Conceptual model
   c) Low-fidelity prototype d) Hierarchical Task Analysis (HTA)
2. What is the main purpose of creating a persona in interaction design?
   a) To list the technical specifications of a database system
   b) To represent a realistic, fictional profile of a core user group
   c) To write step-by-step code for user interface elements
   d) To record analytics from web traffic automatically
3. A series of screen sketches or drawings arranged in sequence to show how a user accomplishes a task is called a:
   a) Persona b) Scenario c) Storyboard d) Web analytics
4. Which evaluation method involves usability experts methodically simulating a user's step-by-step problem-solving process to find interaction flaws?
   a) Cognitive walkthrough b) Web analytics c) Focus groups d) Contextual Inquiry
5. When a system indicates that an option is currently unavailable by graying it out in a menu, it is implementing which design principle?
   a) Affordance b) Feedback c) Consistency d) Constraints
6. Which usability goal focuses on whether a product provides the specific set of functions and tools necessary for users to achieve their objectives?
   a) Safety b) Utility c) Memorability d) Efficiency
7. What type of user evaluation is typically performed early in the design cycle to gather quick, informal feedback on initial ideas?
   a) Thorough laboratory testing b) Field study metrics
   c) Opportunistic evaluation d) Predictive modeling
8. Evaluating whether an interactive product can be physically operated by users who have limited hand or finger dexterity relates to their:
   a) Cognitive capabilities b) Sensory capabilities
   c) Physical capabilities d) Intellectual capabilities
9. Which evaluation technique uses a human designer behind the scenes to manually simulate a system's smart responses before any real software is written?
   a) Heuristic evaluation b) Questionnaire profiling
   c) Wizard of Oz d) Hierarchical Task Analysis (HTA)
10. A narrative description that walks through a realistic sequence of events or user actions under regular or exceptional conditions is a:
    a) Concept model b) Scenario c) Heuristic list d) Prototype layout

PART 2: Short Answer Questions [30 Marks]

Question 1: [10 marks]

a) Contrast the primary functions of personas and scenarios in user experience design. How do they work together? [4 Marks]

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b) A university is building a mobile portal. Briefly outline key traits you would include to draft distinct personas for:
    - A first-year undergraduate student registering for classes. [3 Marks]
    - An academic advisor monitors student progress. [3 Marks]

Question 2: [10 marks]
Identify the core interaction design principle being described or implemented in each of the following scenarios: [2.5 Marks each]

a) A web application changes the mouse cursor into a small hand pointer whenever it hovers over a clickable image link.
b) A smart thermostat screen flashes a checkmark icon and plays a chime sound immediately after a user updates the target room temperature.
c) An e-commerce mobile application utilizes the same layout grid, icon style, and button coloring on its checkout page as it does on its shopping cart page.
d) A digital audio system prevents a user from accidentally activating the "Format Storage" function by hiding it completely inside an advanced settings sub-menu.

Question 3: [10 marks]

a) Compare Contextual Inquiry and Usability Testing based on where the evaluation happens and how data is gathered. [6 Marks]
b) State which of these two methods is most appropriate if an interactive designer wants to discover the messy, real-world workplace disruptions that office clerks face while using data entry software. Explain why. [4 Marks]

PART 3: Long/Problem Solving Questions [50 Marks]

Question 1: [15 marks]

a) Define the core functional differences between a low-fidelity prototype and a high-fidelity prototype. Provide two concrete examples of low-fidelity techniques. [6 Marks]
b) What is a conceptual model in interaction design? List the four structural components that designers typically map out when constructing a conceptual model for a new interface. [9 Marks]

Question 2: [20 marks]
Consider a digital online platform designed for telemedicine appointments, where patients can schedule consultations and talk directly with doctors via live video.

a) Identify two distinct physical environments or contexts where patients are likely to access this telemedicine application, and list two types of computing hardware devices they would use. [6 Marks]
b) List three essential interface elements or specific user tasks that a patient must frequently interact with while navigating through this telehealth platform. [6 Marks]
c) Propose four distinct user experiences or usability goals that design teams must achieve to ensure that elderly or non-technical patients find the platform easy and comfortable to use. [8 Marks]

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Question 3: [15 marks]

a) Imagine you are evaluating a new web portal and need to verify its suitability for users with accessibility needs. Explain how your approach would differ if you were tasked with:
   - Conducting a quick, low-cost preliminary check for visual impairment compatibility. [3 Marks]
   - Executing a highly thorough, representative evaluation of visual impairment suitability. [3 Marks]

b) If an evaluation participant is completely unable to speak or hear, describe one practical accommodation or tool an evaluator could use to track their immediate satisfaction during an interactive task. [3 Marks]

c) Interaction designers must protect the rights and well-being of participants during evaluation studies. Identify and explain two critical ethical responsibilities or legal guidelines (such as informed consent or data privacy) that an evaluator must follow. [6 Marks]

End of questions

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